A SURJECTION OF QUASI-COMPACT HAUSDORFF SPACES OVER AN EXTREMALLY DISCONNECTED TARGET WHICH IS MINIMAL IS A HOMEOMORPHISM

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Allowed standard prerequisites (editorial).

- The statement that a bijective continuous map from a quasi-compact to a Hausdorff space is a homeomorphism. Invoked at the start of the counted proof (source line 5488). Reduces the assertion to injectivity.
- The statement that a continuous map of quasi-compact spaces with extremally disconnected target is closed. Invoked inside the counted proof (source lines 5492-5493). Makes the images of the two complements closed.
- The technical containment for the image of an open subset. Invoked inside the counted proof (source lines 5496-5498). Places the common image point in the closures of the two complements.
- The statement that closures of disjoint opens in an extremally disconnected space are disjoint. Invoked at the end of the counted proof (source lines 5498-5499). Contradicts the two closure memberships.

Source reference. Original lemma lemma-bijective-map: sources/stacks-topology/pinned-original-source.tex, source lines 3498–3498. Navigation pointer only: the counted text refers to the Stacks item lemma-bijective-map, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-closed-map: sources/stacks-topology/pinned-original-source.tex, source lines 3474–3474. Navigation pointer only: the counted text refers to the Stacks item lemma-closed-map, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the

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Source reference. Original lemma lemma-image-open-technical: sources/stacks-topology/pinned-original-source.tex, source lines 5449–5449. Navigation pointer only: the counted text refers to the Stacks item lemma-image-open-technical, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-intersection-empty: sources/stacks-topology/pinned-original-source.tex, source lines 5466–5466. Navigation pointer only: the counted text refers to the Stacks item lemma-intersection-empty, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $f : X \to Y$ be a continuous map of Hausdorff quasi-compact topological spaces. If Y is extremally disconnected, f is surjective, and $f(Z) \neq Y$ for every proper closed subset Z of X, then f is a homeomorphism.*

Proof. By Lemma lemma-bijective-map it suffices to show that f is injective. Suppose that $x, x' \in X$ are distinct points with $y = f(x) = f(x')$. Choose disjoint open neighbourhoods $U, U' \subset X$ of x, x'. Observe that f is closed (Lemma lemma-closed-map) hence $T = f(X \setminus U)$ and $T' = f(X \setminus U')$ are closed in Y. Since X is the union of $X \setminus U$ and $X \setminus U'$ we see that $Y = T \cup T'$. By Lemma lemma-image-open-technical we see that y is contained in the closure of $Y \setminus T$ and the closure of $Y \setminus T'$. On the other hand, by Lemma lemma-intersection-empty, this intersection is empty. In this way we obtain the desired contradiction. □